A-A-2555
March 17, 1988

COMMERCIAL ITEM DESCRIPTION

DRILL, MASONRY, HAND
(STAR (FOUR-FLUTE) DRILL)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-D-691, Type I.

This commercial item description covers star masonry hand drills for manual use.

SALIENT CHARACTERISTICS

Design. The cutting end shall have four cutting edges and shall have an equal number of flutes. The cutting face shall be slightly pointed and edges shall be sharp. The striking end shall be slightly chamfered. The body shall be either octagonal or hexagonal in cross section.

Material. All drills shall be forged from a high quality, wear- and shock-resistant steel conforming to SAE 9255 or AISI 9255. The drills shall be hardened and tempered so that they will withstand the tests specified.

Finish. Drills shall have a lacquered, enameled, or oil-treated natural finish at the option of the contractor.

Sizes. See Table I.

Table I. Star (four - flute) drill

National Stock Number	Nominal Size Inches	Diameter cutting end Min Inches	Diameter cutting end Max Inches	Stock size Min Inches	Length fluted section (Min) Inches	Length overall (approx) Inches
5118-00-246-6141	1/4	0.250	0.281	3/16	1-1/4	12
5118-00-236-2847	3/8	.375	.410	5/16	1-3/8	12
5118-00-186-7185	1/2	.500	.531	3/8	1-3/4	12
5118-00-234-1956	5/8	.625	.680	7/16	2	12
5118-00-221-8134	5/8	.625	.680	7/16	2	18
5118-00-236-2846	3/4	.750	.795	1/2	2-1/4	12
5118-00-223-1185	3/4	.750	.795	1/2	2-1/4	18
5118-00-234-1957	7/8	.875	.921	5/8	2-1/2	12
5118-00-236-2849	1	1.000	1.046	11/16	2-1/2	12
5118-00-223-7341	1	1.000	1.046	11/16	2-1/2	18
5118-00-186-7899	1-1/8	1.125	1.171	3/4	2-3/4	12
5118-00-234-1923	1-3/4	1.750	1.812	1-1/8	3-5/8	36

Identification marking. Each drill shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number, and country of origin.

Workmanship. All drills shall be free from imperfections which may affect their safety, appearance, serviceability, or durability.

FSC 5118

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QUALITY ASSURANCE PROVISIONS

Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes, except where otherwise specified herein. The lot size shall consist of all drills offered for acceptance at one time. The sample unit is one drill.

Visual and dimensional inspection. Each drill from the sample lot shall be examined for workmanship, marking, and dimensional requirements. The inspection level shall be I with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Testing of end item. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

Hardness test. The cutting end shall be hardened and tempered to a Rockwell C hardness of not less than 52 nor more than 60. Impressions shall be taken within a distance of 1/4 to 1/2 inch back of the cutting edge. Hardness shall be in accordance with ASTM E18.

Drilling test. The drill shall be supported with the cutting end against a solid concrete block. The striking end of the drill shall be struck with a hammer of weight as specified as below until a hole of not less than 1/2 inch in depth has been attained. A minimum of three holes shall be drilled. There shall be no indication of bending, deforming, or fracture in the body of the drill, nor any signs of spalling, chipping, or cracking of the cutting end. Failure of drill to perform satisfactorily shall be cause for rejection.

For drill sizes 1/4 through 7/16 inclusive, use 1 pound hammer.

For drill sizes 1/2 through 5/8 inclusive, use 1-1/2 pound hammer.

For drill sizes 11/16 through 1 inclusive, use 2 pound hammer.

For drill sizes 1-1/8 through 1-3/4 inclusive, use 3 pound hammer.

Marking test. Following the drilling tests, drills shall be subjected to the marking test. Using a hammer of the proper weight as specified in the drilling test, three impressions shall be made with the cutting end of the drill on a steel plate having a Rockwell C hardness of not less than 37 nor more than 48. When tested as specified in the drilling and marking test, the striking end of the drill shall not mushroom to the extent that cracks develop on the edge. There shall be no indication of bending, deforming, or fracture in the body of the drill, nor any signs of spalling, chipping, or cracking of the cutting end. Failure of drill to perform satisfactorily shall be cause for rejection.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

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Notes.

The issue of referenced documents in effect on date of invitation for bid or request for proposal shall be used to determine compliance with the stated requirements.

AISI standards are available from American Iron and Steel Institute, 1133 15th Street NW, Suite 300, Washington, DC 20005.

ASTM standards are available from American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

SAE standards are available from Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.

Military standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MILITARY INTERESTS:

Military Coordinating Activity
Army - GL

Custodians:
Army - GL
Air Force - 99

Review Activities:
Air Force - 84

PREPARING ACTIVITY:

GSA/FSS

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